

# Matheus Leca

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Calgary, Alberta

## Education

- **M.Sc., Electrical and Computer Engineering, University of Calgary, CAN** Graduated 2026
- **B.Sc., Computer Science, Federal University of Pernambuco, BRA** Graduated 2024

## Work History

*Software Engineer, University of Calgary, CAN*

*Apr 2025 – Aug 2026*

- Led the development of a cross-platform safety-training simulator using **Unity 6, C#, HTML, CSS, and JavaScript** to train employees to identify workplace hazards through immersive 3D and 360° scenarios across web, mobile, desktop, and Meta Quest VR.
- Built an online training backend using **.NET, EF Core, and PostgreSQL** to manage training level creation and availability, sharing, leaderboards, and employee submissions with **Clean Architecture**.
- Automated deployments, database migrations, and testing using **Git, GitHub, GitHub Actions, Docker, and DigitalOcean**, applying **Test-Driven Development** to support reliable delivery.
- Engineered a custom **Unity WebGL asset pipeline** with automated asset splitting, parallel downloads, and client-side reassembly to overcome the 100 MB hosting limit and enable deployment of the full simulator.

*Software Engineer, LogAp I.T. Solutions, BRA*

*Feb 2023 – Aug 2024*

- Led the development of an enterprise application using **Java, Spring Boot, microservices, Angular, and Kafka** to manage inventory and predict repositioning needs, streamlining government workflows and reducing operational costs while collaborating with stakeholders to refine requirements and deliver requested features.
- Digitized electricity-service operations using **Ionic, REST APIs, RabbitMQ, and AWS** to monitor service availability and requests and automate asynchronous data exchange, improving operational efficiency and supporting revenue growth.
- Developed and optimized a flood-monitoring application using **C#, .NET, HTML, CSS, and JavaScript** to collect real-time water-level data and support emergency-service responses.
- Improved application maintainability by applying **SOLID principles** and supported reliable delivery using **Git, GitHub, and GitHub Actions** for automated CI/CD and testing across **Linux and Unix** environments throughout the **Software Development Life Cycle (SDLC)**, following **Agile/Scrum** practices and conducting code reviews.

*Software Engineer, Reciprev, BRA*

*Oct 2021 – Feb 2023*

- Developed **React, Vue, jQuery, HTML, CSS, and JavaScript** workflows to digitize and streamline social-security accounting processes, reducing software costs by collaborating with stakeholders to refine functional and non-functional requirements.
- Enabled reliable cloud deployment using **Node.js, Express, MongoDB, and AWS**, supporting scalable application hosting, backend services, and persistent application data.
- Improved software reliability using **Git, GitLab, Test-Driven Development, Agile/Scrum** and code reviews throughout the SDLC, supporting consistent and reliable feature delivery.

*Teaching Assistant, University of Calgary, CAN*

*Sep 2024 – Apr 2026*

- Improved students' understanding of software architecture and design patterns using **Java, Spring, Spring Boot, HTML, CSS, and JavaScript** through lectures and practical instruction on microservices and modular monoliths.
- Strengthened students' **object-oriented programming and data structures** skills using **Python, MySQL, and MongoDB** by providing individualized guidance and feedback on labs, assignments, and projects.
- Supported **programming, databases, software design, architecture, and web development** instruction across six undergraduate and graduate courses, helping students develop practical skills in these areas.

Teaching Assistant, Federal University of Pernambuco, BRA

Jul 2022 – Mar 2024

- Enhanced undergraduate learning using **JavaScript, TypeScript, Node.js, and OpenAI Assistants API** to develop an AI tutoring application integrated with Discord, providing accessible, on-demand academic support.
- Strengthened students' understanding of **Theoretical Foundations of Computer Science** through individualized explanations, feedback, and academic guidance.
- Improved learning and assessment by delivering lectures, creating and grading coursework, and holding weekly office hours to explain technical concepts in plain language and provide individualized support.

## Projects

VeriFin, Personal Web Application

2026

- Developed a full-stack personal finance platform using **Java, Spring Boot, React, TypeScript, Tailwind CSS, and PostgreSQL** to help users organize budgets, track expenses, set financial goals, and monitor investments in one application.

ThriveTree, Personal iOS Project

2026

- Developed a **Swift and SwiftUI** iOS app that gamifies habit building through a virtual tree, using animations, reminders, health states, and progress tracking to encourage consistent daily habits.

## Publications

- *FairPath: A Human-in-the-Loop Framework for Adversarial Fairness Testing in Machine Learning Pipelines.* Master's Thesis, 2026.
- *A Systematic Review of Bias Detection and Dataset Improvement Methods for Fair AI Systems.* FSE (HumanAISE '26), 2026.
- *Empathy in Software Engineering Education: Evidence, Practices, and Opportunities.* ICSE (CHASE '26), 2026.
- *Software Fairness Testing in Practice.* ICSME '25, 2025.
- *Curious, Critical Thinker, Empathetic, and Ethically Responsible: Essential Soft Skills for Data Scientists in Software Engineering.* ICSE (SEIS '25), 2025.
- *Towards User-Focused Cross-Domain Testing: Disentangling Accessibility, Usability, and Fairness.* SANER (SQA4AI '25), 2025.
- *Responsible AI in the Software Industry: A Practitioner-Centered Perspective.* ICSE (RAIE '25), 2025.
- *Applications and Implications of Large Language Models in Qualitative Analysis: A New Frontier for Empirical Software Engineering.* ICSE (WSESE '25), 2025.
- *Development and Evaluation of an Artificial Intelligence-Based Chatbot as a Learning Support Tool.* Bachelor's Thesis, 2023.